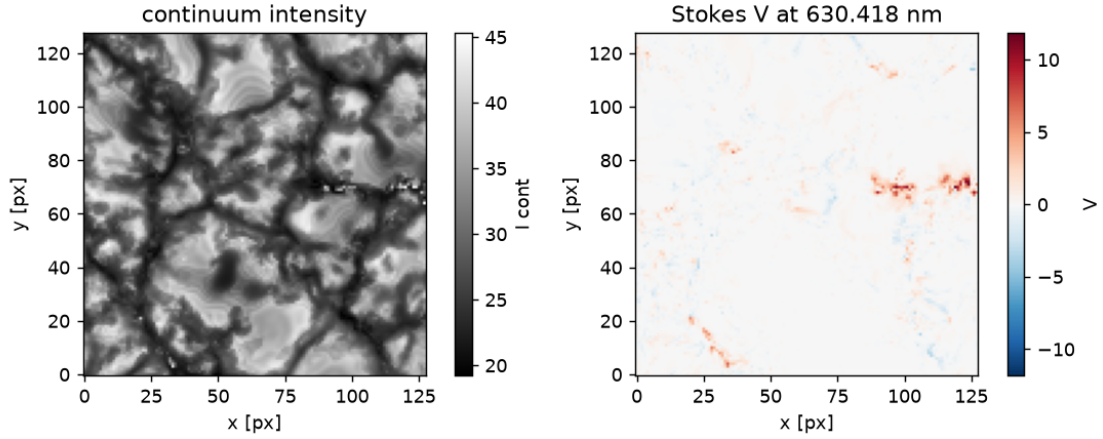


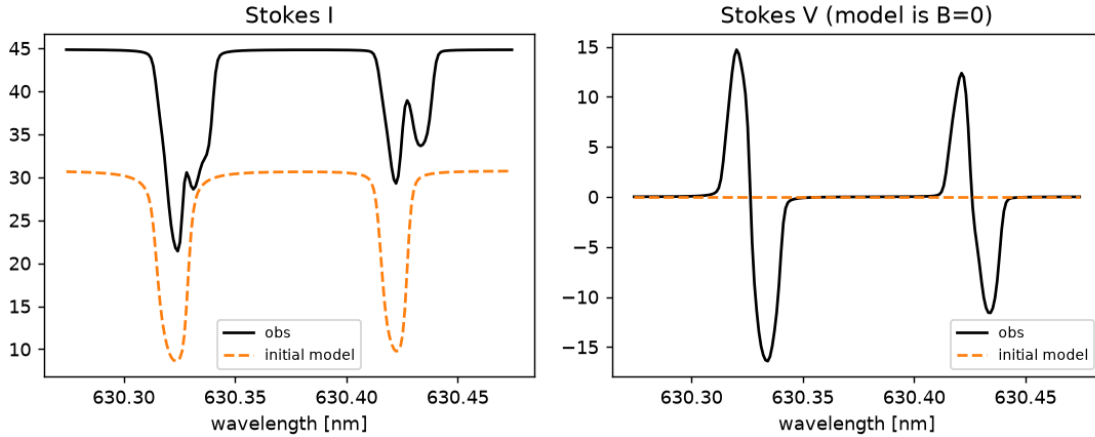
Adora PINN — Step 5: Mock inversion testset

As-built notes · Ivan Milic · 2026-09-20 · step 5 of pinn/plan.md

The mock inversion testset — the Stokes observations the PINN will invert — is a crop of a prepared synthetic cube, [/dat/milic/adora_synth/test.npz.npy](#) (1536×1536×4×201): the first 384×384 pixels, every 3rd, giving 128×128 **pixels at 48 km horizontal spacing** (the original 16 km ×3; it will matter for 3D hydrostatic equilibrium later). It is a magnetized MURaM field (V up to ± 16 , granular continuum), synthesized with the same Adora RT as the forward model, so its Stokes units and wavelength grid (Fe I 6301/6302, 201 vac points) match the PINN's directly. **Saved to** [/dat/milic/adora_pinn_develop/inversion_testset.npz](#) (keys `I_obs`, `waves`, `dx_km=dy_km=48`). Against it, the initial model (the pretrained FAL-C field, non-magnetic) has misfit $\text{RMS}(\text{model} - \text{obs})/\langle I_c \rangle$ of I 0.20, Q 1.2×10^{-3} , U 1.2×10^{-3} , V 9.3×10^{-3} — V entirely missing since the model has $B=0$ — which is the signal the step-6 losses will drive down. Builder [pinn/make_inversion_testset.py](#); notebook [pinn/step5_testset.ipynb](#).



Observations: continuum granulation (left) and the Stokes V polarity map (right).



Initial model vs. observations at the most strongly polarised pixel: Stokes I differs and Stokes V is absent in the non-magnetic starting model — the starting point of the inversion.